

**CLASSROOM CONTACT PROGRAMME**
(Academic Session : 2019 - 2020)

NEET : PRACTICE PAPER

TARGET : PRE-MEDICAL - 2019

Test Type : **PT # 02**

Test Pattern : NEET

TEST DATE : 01-04-2020

Important Instructions

Do not open this Test Booklet until you are asked to do so

1. A seat marked with Reg. No. will be allotted to each student. The student should ensure that he/she occupies the correct seat only. If any student is found to have occupied the seat of another student, both the students shall be removed from the examination and shall have to accept any other penalty imposed upon them.
2. Duration of Test is **3 Hours** and Questions Paper Contains **180** Questions. The Max. Marks are **720**.
3. Student can not use log tables and calculators or any other material in the examination hall.
4. Student must abide by the instructions issued during the examination, by the invigilators or the centre incharge.
5. Before attempting the question paper ensure that it contains all the pages and that no question is missing.
6. Each correct answer carries 4 marks, while **1 mark will be deducted for every wrong answer**. Guessing of answer is harmful.
7. A candidate has to write his / her answers in the OMR sheet by darkening the appropriate bubble with the help of **Blue / Black Ball Point Pen only** as the correct answer(s) of the question attempted.
8. **Use of Pencil is strictly prohibited.**

Note : In case of any Correction in the test paper, please mail to dlpcorrections@allen.ac.in within 2 days along with **Paper code** and Your **Form No.**

Your Target is to secure Good Rank in Pre-Medical

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TOPIC COVER

PHYSICS : Motion in 1D + Projectile Motion + Relative Motion + Laws of Motion + Friction

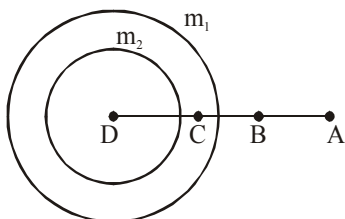
CHEMISTRY : Thermodynamics and Energetics (Complete) + Solid state + Chemical and Ionic Equilibrium + Isomerism

BIOLOGY : Genetics + Biotech + DOP (Complete) + Ecology + Plant reproduction(Complete) + Protoplasm

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

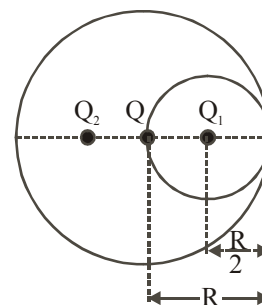
BEWARE OF NEGATIVE MARKING

- If the earth stops rotating suddenly the value of g at a place other than poles would :-
(1) Decrease
(2) Remains constant
(3) Increase
(4) Increase or decrease depending on the position of earth in the orbit round the sun
- What should be the angular velocity of earth about own axis so that a persons weight at equator will be $3/5$ of his weight at poles :-
(1) $\sqrt{\frac{g}{R}}$ (2) $\sqrt{\frac{2g}{3R}}$ (3) $\sqrt{\frac{2g}{5R}}$ (4) $\sqrt{\frac{3g}{2R}}$
- The depth from the surface of the earth of radius R at which the acceleration due to gravity will be 75% of the value of the surface of the earth is :-
(1) $\frac{R}{4}$ (2) $\frac{R}{2}$ (3) $\frac{R}{8}$ (4) $\frac{3R}{4}$
- A body has mass of 1 kg on the surface of the Earth. What is the mass at the centre of the Earth
(1) Zero (2) 1kg (3) 9.8 kg (4) ∞
- Figure shows two shells of masses m_1 and m_2 . The shells are concentric. At which point, a particle of mass m should experience zero force:-



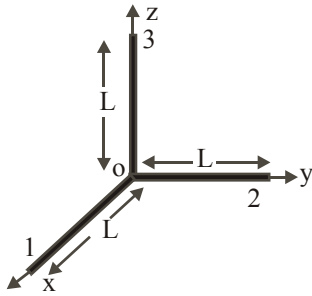
- (1) A (2) B (3) C (4) D
- A body of mass M is dropped from a height h on a sand floor. If the body penetrates x cm into the sand, the average resistance offered by the sand to the body is :-
(1) $Mg\left(\frac{h}{x}\right)$ (2) $Mg\left(1 + \frac{h}{x}\right)$
(3) $Mgh + Mg x$ (4) $Mg\left(1 - \frac{h}{x}\right)$

- Force acting on a particle moving in a straight line varies with the velocity of the particle as $F = \frac{K}{v}$, where K is a constant. The work done by this force in time t is :-
(1) $\frac{K}{v^2}t$ (2) $2Kt$ (3) Kt (4) $\frac{2Kt}{v^2}$
- A mass m moving horizontally with velocity v_0 strikes a pendulum of mass m . If the two masses stick together after the collision, then the maximum height reached by the pendulum is :-
(1) $v_0^2/8g$ (2) $v_0^2/2g$
(3) $\sqrt{2v_0g}$ (4) $\sqrt{v_0g}$
- When two bodies collide elastically, then :-
(1) KE of the system alone is conserved
(2) only momentum is conserved
(3) both KE and momentum are conserved
(4) neither KE nor momentum is conserved
- A body falls from a height of 16 m and rebounds to a height of 4m. The coefficient of restitution is
(1) $\frac{1}{4}$ (2) $\frac{1}{2}$ (3) $\frac{3}{4}$ (4) $\frac{1}{8}$
- A spherical hollow is made in a lead sphere of radius R , such that its surface touches the outside surface of lead sphere and passes through the centre. What is the shift in the centre of mass of lead sphere due to the hollowing :-



- (1) $\frac{R}{7}$ (2) $\frac{R}{14}$ (3) $\frac{R}{2}$ (4) R

12. Three thin rods of length L and mass M are placed along x , y and z -axes in such a way that one end of each of the rods is at the origin. The moment of inertia of this system about z -axis is :-

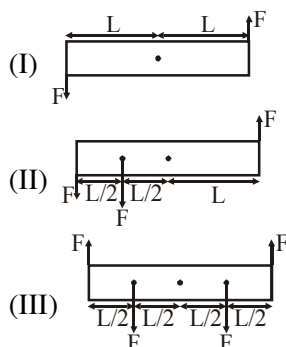


- (1) $\frac{2ML^2}{3}$ (2) $\frac{4ML^2}{3}$ (3) $\frac{5ML^2}{3}$ (4) $\frac{ML^2}{3}$
13. A wire of mass m and length ℓ is bent in the form of circular ring. The moment of inertia of the ring about its axis is :-
- (1) $m\ell^2$ (2) $\frac{m\ell^2}{4\pi^2}$ (3) $\frac{m\ell^2}{2\pi^2}$ (4) $\frac{m\ell^2}{8\pi^2}$
14. A ring is rolling on a surface without slipping. What is the ratio of its translation to rotational kinetic energies :-
- (1) 5 : 7 (2) 2 : 5 (3) 2 : 7 (4) 1 : 1
15. The potential energy for a conservative system is given by:

$$U = ax^2 - bx$$

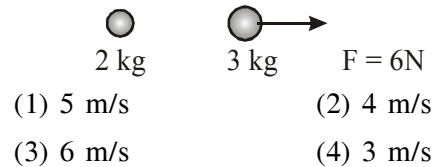
Where a and b are positive constants. The law of the force governing the system is

- (1) $F = \text{const}$ (2) $F = bx - 2a$
(3) $F = b - 2ax$ (4) $F = 2ax$
16. A rigid rod of length $2L$ is acted upon by some forces. All forces labelled F have the same magnitude. Which cases have a non-zero net torque acting on the rod about its centre ?

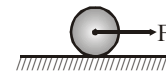


- (1) I and II only
(2) II and III only
(3) I and III only
(4) The net torque is zero in all cases

17. Two particles are shown in Fig. At $t = 0$, a constant force $F = 6 \text{ N}$ starts acting on the 3 kg mass. The velocity of the centre of mass of these particles at $t = 5 \text{ s}$.



- (1) 5 m/s (2) 4 m/s
(3) 6 m/s (4) 3 m/s
18. A uniform disc of mass M radius R , is resting on a table on its rim. The coefficient of friction between disc and table is μ . Now the disc is pulled with a force F as shown in the figure. What is the maximum value of F for which the disc rolls without slipping ?



- (1) μMg (2) $2\mu Mg$
(3) $3\mu Mg$ (4) $4\mu Mg$
19. Two discs of moments of inertia I_1 and I_2 about their respective axes, rotating with angular frequencies ω_1 and ω_2 in same sense respectively, are brought into contact face to face with their axes of rotation coincident. The angular frequency of the composite disc will be :-

- (1) $\frac{I_1\omega_1 + I_2\omega_2}{I_1 + I_2}$ (2) $\frac{I_2\omega_1 + I_1\omega_2}{I_1 + I_2}$
(3) $\frac{I_1\omega_1 - I_2\omega_2}{I_1 - I_2}$ (4) $\frac{I_2\omega_1 - I_1\omega_2}{I_1 - I_2}$

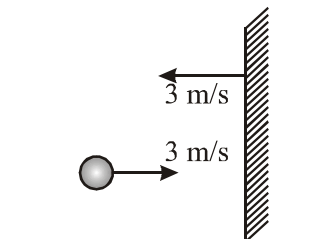
20. 300 J of work done in sliding a 2kg block up an inclined plane of height 10m. Taking $g = 10 \text{ ms}^{-2}$, work done against friction is :-

- (1) 1000 J (2) zero
(3) 200 J (4) 100 J

21. A solid cylinder of mass M and radius R rolls without slipping down an inclined plane making an angle θ with the horizontal. Then its acceleration is :-

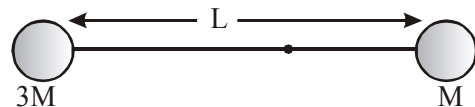
- (1) $\frac{1}{3}g\sin\theta$ (2) $\frac{2}{3}g\sin\theta$
(3) $\frac{2}{5}g\sin\theta$ (4) $\frac{7}{7}g\sin\theta$

22. The kinetic energy K of a particle moving in a straight line depends upon the distance s as; $K = as^2$. The force acting on the particle is :-
 (1) $2as$ (2) $2mas$ (3) $2a$ (4) $\sqrt{as^2}$
23. A body of mass m_1 moving with uniform velocity of 40 m/sec, collides with another mass m_2 at rest and then the two together begin to move with a uniform velocity of 30 m/sec. The ratio of their masses (m_1/m_2) is :-
 (1) 0.75 (2) 1.33 (3) 3.0 (4) 4.0
24. The moment of inertia of a circular ring of radius R and mass M about a tangent in its plane is:-
 (1) MR^2 (2) $(1/2)MR^2$
 (3) $(3/2)MR^2$ (4) $2MR^2$
25. The speed of a homogeneous solid sphere after rolling down an inclined plane of vertical h , from rest, without sliding is :-
 (1) $\sqrt{gh}$ (2) $\sqrt{(6/5)gh}$
 (3) $\sqrt{(4/3)gh}$ (4) $\sqrt{(10/7)gh}$
26. Three thin metal rods, each of mass M and length L , are welded to form an equilateral triangle. The moment of inertia of the composite structure about an axis passing through the centre of mass of the structure and perpendicular to its plane is :-
 (1) $\frac{1}{2}ML^2$ (2) $\frac{1}{3}ML^2$
 (3) $\frac{2}{3}ML^2$ (4) $\frac{1}{4}ML^2$
27. A highly elastic ball moving at a speed of 3 m/s approaches a wall moving towards it with a speed of 3 m/s. After the collision, the speed of the ball will be :



- (1) 3 m/s (2) 6 m/s
 (3) 9 m/s (4) Zero

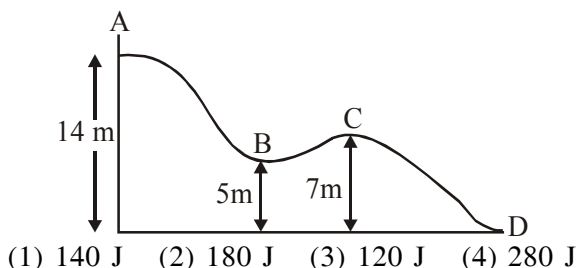
28. Two cars of masses m_1 and m_2 are moving in circles of radii r_1 and r_2 respectively. Their speeds are such that they make complete circles in the same time t . The ratio of their centripetal acceleration is :-
 (1) 1 : 1 (2) $m_1r_1 : m_2r_2$
 (3) $m_1 : m_2$ (4) $r_1 : r_2$
29. Three bodies, a ring, a solid cylinder and a solid sphere roll down the same inclined plane without slipping. They start from rest. The radii of the bodies are identical. Which of the bodies reaches the ground with maximum velocity ?
 (1) Ring
 (2) Solid cylinder
 (3) Solid sphere
 (4) All reach the ground with same velocity
30. Two masses M and $3M$ are connected by a string of length L rotated in a horizontal plane as shown in figure find out distance of axis from M so that tension in both part of string is same.



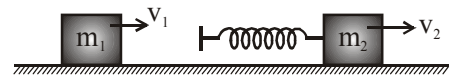
- (1) $\frac{3L}{4}$ (2) $\frac{L}{4}$ (3) $\frac{L}{2}$ (4) $\frac{3L}{5}$
31. A particle of mass 200 g is dropped from a height of 50 m and another particle of mass 100 g is simultaneously projected up from the ground along the same line with a speed of 100 m/s. The acceleration of the centre of mass after 1 s is :
 (1) 10 m/s^2 (2) $\frac{10}{3} \text{ m/s}^2$
 (3) 0 (4) none of these
32. A car of mass 123 Kg negotiates a banked curve of radius 120 m on a frictionless road. If the banking angle is 37° the speed of the car is :
 (1) 5 m/s (2) 10 m/s
 (3) 20 m/s (4) 30 m/s
33. Three identical rods, each of length L , are joined to form a rigid equilateral triangle. Its radius of gyration about an axis passing through a corner and perpendicular to plane of the triangle is :-

- (1) $\frac{L}{2}$ (2) $\sqrt{\frac{3}{2}}L$ (3) $\frac{L}{\sqrt{6}}$ (4) $\frac{L}{\sqrt{2}}$

34. Which of the following statement is correct :-
 (1) Parallel axis theorem is valid only for 2-D bodies
 (2) Perpendicular axis theorem is valid for all types of bodies
 (3) Parallel axis theorem and perpendicular axis theorem are applicable for all types of bodies
 (4) Perpendicular axis theorem is valid for 2-D bodies and parallel axis theorem is valid for all types of bodies
35. A particle of mass 1kg is kept at (1m, 1m, 1m). The moment of inertia of this particle about z-axis would be :-
 (1) 1 kg-m² (2) 2kg-m²
 (3) 3 kg-m² (4) None of these
36. Two loops P and Q are made of uniform wire. The radii of P and Q are R₁ and R₂ respectively and their moment of inertia are I₁ and I₂ respectively. If $\frac{I_2}{I_1} = 5$ then $\frac{R_2}{R_1}$ is :-
 (1) 5^{2/3} (2) 5^{1/3} (3) 5^{-2/3} (4) 5^{-1/3}
37. If a number of forces act on a body and the body is in static equilibrium, then :-
 (1) Work done by any individual force must be zero
 (2) Net work done by all the forces is positive
 (3) Net work done by all the forces is negative
 (4) Net work done by all the forces is zero
38. A mass M is lowered with the help of a string by a distance h at a constant acceleration g/2. The work done by the string will be:-
 (1) $\frac{Mgh}{2}$ (2) $-\frac{Mgh}{2}$
 (3) $\frac{3Mgh}{2}$ (4) $-\frac{3Mgh}{2}$
39. Fig. shows the vertical section of a frictionless surface. A block of mass 2 kg is released from rest from position A; its KE as it reaches position C is (g = 10 ms⁻²):-

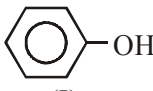
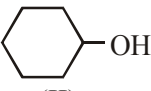
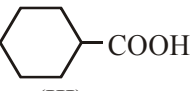
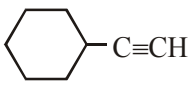


40. The momentum of a body is increased by 50%. The K.E. of the body will be increased by
 (1) 50% (2) 125%
 (3) 330% (4) 400
41. For a vertical moving on banked road of radius of curvature r and angle of banking 'θ', the speed for which no wear or tear occurs to tyres is :-
 (1) $v \leq \sqrt{rg \tan \theta}$ (2) $v \geq \sqrt{rg \tan \theta}$
 (3) $v \leq \sqrt{\frac{rg \tan \theta}{2}}$ (4) $v = \sqrt{rg \tan \theta}$
42. Two masses m₁ = 2kg and m₂ = 5 kg moving on a frictionless surface with velocities 10 m/s and 3m/s respectively, m₂ is ahead of m₁. An ideal spring of spring constant K = 7000 N/m is attached on the back side of m₂. The maximum compression of the spring will be :-

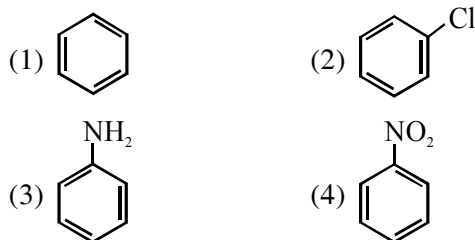


- (1) 0.51 m (2) 0.062 m
 (3) 0.1 m (4) 0.72 m
43. A ball is dropped from a height of 8.1m on a floor, after 2nd rebound it rises to a height of 0.1m above the floor. What is coefficient of restitution between the ball and the floor:-
 (1) $\frac{1}{3}$ (2) $\frac{1}{9}$
 (3) $\frac{1}{27}$ (4) None of these
44. The distance between two spheres of mass m₁ and m₂ is r. Their distance from the C.M. are r₁ and r₂ respectively. Then
 (1) $\frac{m_1}{m_2} = \frac{r_2}{r_1}$ (2) $\frac{m_1}{m_2} = \frac{r_1}{r_2}$
 (3) $\frac{m_1}{r_1^2} = \frac{m_2}{r_2^2}$ (4) m₁m₂ = r₁r₂
45. Two particles of mass m₁ and m₂ are projected from the top of a tower. The particle m₁ is projected vertically downward with speed u and m₂ is projected horizontally with same speed then acceleration of centre of mass of system of particles neglecting air resistance is :

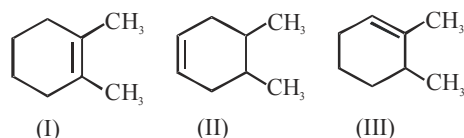
- (1) $\sqrt{2} g$ (2) g (3) $\frac{g}{\sqrt{2}}$ (4) $\frac{g}{2}$

- 46.** Inductive effect involves :-
 (1) Partial displacement of π -electrons
 (2) Delocalisation of π -electrons
 (3) Partial displacement of σ (sigma) electrons
 (4) None
- 47.** Arrange the following compound in the basic strength order ?
 (1) $\text{CH}_3\text{-CH}_2\text{-NH}_2$
 (2) $\text{CH}_3\text{-CH=}\ddot{\text{N}}\text{H}$
 (3) $\text{CH}_3\text{-C}\equiv\ddot{\text{N}}$
 (1) $3 > 2 > 1$ (2) $2 > 3 > 1$
 (3) $1 > 2 > 3$ (4) $1 > 3 > 2$
- 48.** Which amongst the following is the most stable carbocation :-
 (1) CH_3CH_2^+ (2) CH_3^+
 (3) $\text{CH}_3\text{-C}^+(\text{CH}_3)_2$ (4) $\text{CH}_3\text{-CH}^+(\text{CH}_3)_2$
- 49.** Which of the following intermediate are electrophilic in character ?
 Carb-cation (a) Carbanion (b) Free radical (c) Carbene (d)
 (1) Only b (2) a and b
 (3) a, c and d (4) a, b, c and d
- 50.** Give the correct order of increasing acidity of the following compounds :-
 (I)  (II)  (III)
 (IV)
 (1) $\text{II} < \text{I} < \text{III} < \text{IV}$
 (2) $\text{IV} < \text{II} < \text{I} < \text{III}$
 (3) $\text{I} < \text{II} < \text{IV} < \text{III}$
 (4) $\text{IV} < \text{I} < \text{II} < \text{III}$

- 51.** Which of the following benzen ring is most reactive towards E.S.R. :-

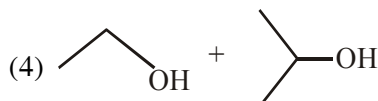
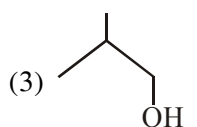
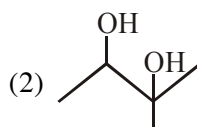
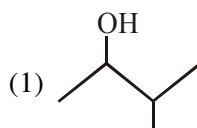


52.

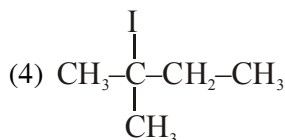
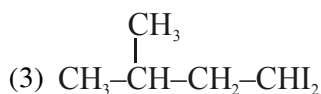
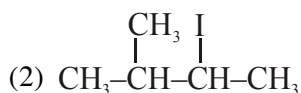
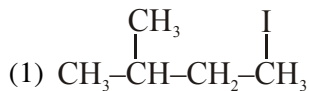


Which of following orders is correct for heat of hydrogenation of these compounds ?

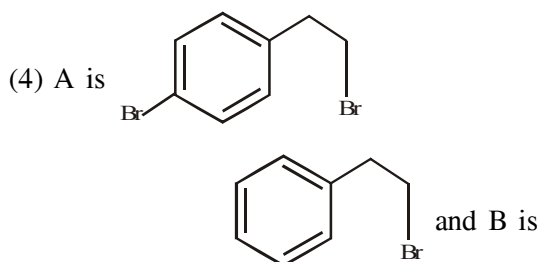
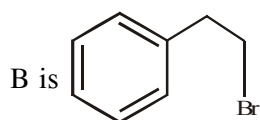
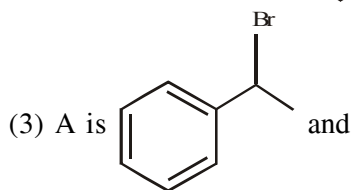
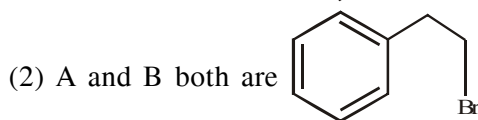
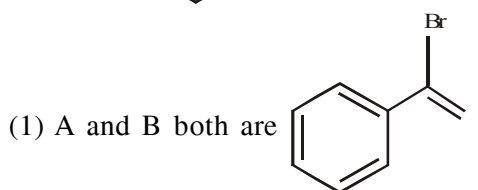
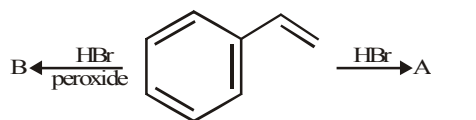
- (1) $\text{I} > \text{II} > \text{III}$
 (2) $\text{III} > \text{II} > \text{I}$
 (3) $\text{II} > \text{III} > \text{I}$
 (4) $\text{III} > \text{I} > \text{II}$
- 53.** $\text{CH}_3\text{-CH=C(CH}_3\text{)-CH}_3 \xrightarrow[\text{H}_2\text{O}_2, \text{OH}^-]{\text{BH}_3, \text{THF}}$ Major product is :-



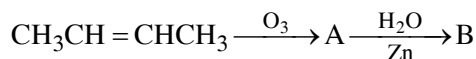
54. Identify major product



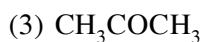
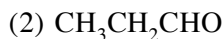
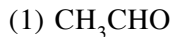
55. Observe the following reactions and predict the nature of A and B.



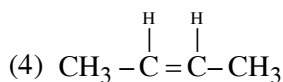
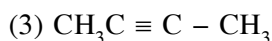
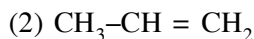
56. In the following sequence of reactions, the alkene affords the compound B



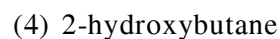
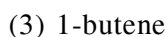
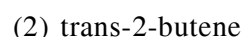
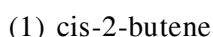
The compound B is



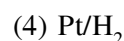
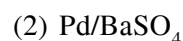
57. The treatment of CH_3MgX with $\text{CH}_3\text{C} \equiv \text{C}-\text{H}$ produces



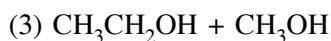
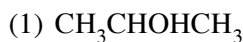
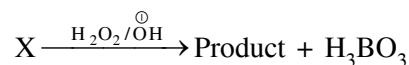
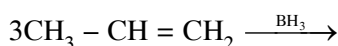
58. When 2-butyne is treated with H_2 in presence of $\text{Pd}-\text{BaSO}_4$; the product formed will be



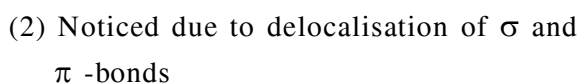
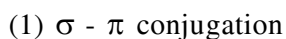
59. 2-Hexyne gives trans-2-hexene on treatment with



60. Identify the product in following reaction



61. Hyper conjugation is



62. The hybridization state of carbocation is -

- (1) sp^3d (2) sp^3 (3) sp^2 (4) sp

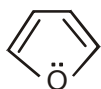
63. Resonance is due to :-

- (1) Delocalization of sigma electrons
(2) Delocalization of pi electrons
(3) Migration of H-atoms
(4) Migration of Protons

64. Number of π -electrons present in the species :



and



would be :

- (1) 2, 4 (2) 1, 4 (3) 2, 6 (4) 1, 6

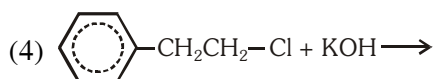
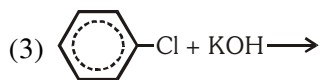
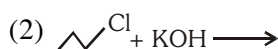
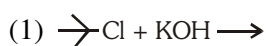
65. Which of the following has maximum pK_a :-

- (1) CH_2FCOOH (2) $CH_2ClCOOH$
(3) CH_3COOH (4) $HCOOH$

66. Aryl halides are less reactive towards nucleophilic substitution reaction as compared to alkyl halides due to -

- (1) The formation of less stable carbonium ion
(2) Resonance stabilization
(3) The inductive effect
(4) sp^3 hybridised carbon attached to the halogen

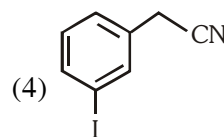
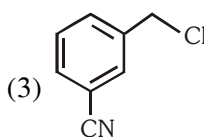
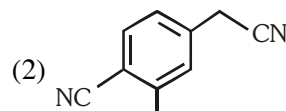
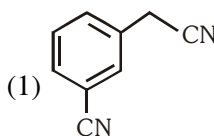
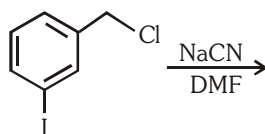
67. SN^1 reaction is feasible in-



68. Tertiary alkyl halide are practically inert to substitution by SN^2 mechanism because of-

- (1) steric hinderance
(2) inductive effect
(3) instability
(4) insolubility

69. The structure of the major product formed in the following reaction is :



70. Reactivity order of halides for dehydrohalogenation is

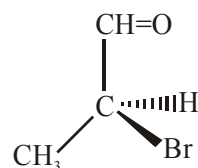
- (1) $R-F > R-Cl > R-Br > R-I$
(2) $R-I > R-Br > R-Cl > R-F$
(3) $R-I > R-Cl > R-Br > R-F$
(4) $R-F > R-I > R-Br > R-Cl$

71. The correct reactivity order of alcohols towards $H-X$ will be

- (I) $CH_2=CH-OH$
(II) $CH_2=CH-CH_2OH$
(III) CH_3-CH_2-OH
(IV) $CH_3-\underset{\text{OH}}{\underset{|}{CH}}-CH_3$

- (1) $II > I > III > IV$ (2) $IV > III > II > I$
(3) $II > IV > I > III$ (4) $II > IV > III > I$

72. The configuration of given compound is :-

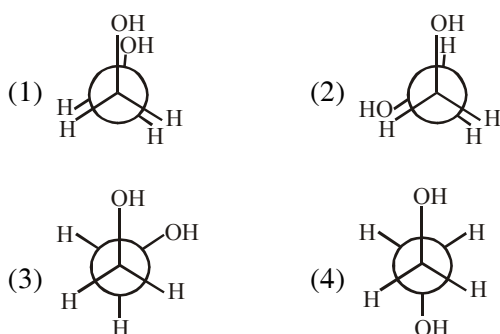


- (1) S (2) Z (3) E (4) R

73. Which of the following statement is true :-

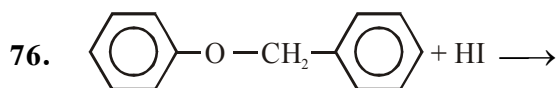
- (1) Trans-But-2-ene is more stable than Cis-But-2-ene
- (2) Cis-But-2-ene has boiling point higher than trans-But-2-ene
- (3) Cis-But-2-ene has melting point lower than trans-But-2-ene
- (4) All

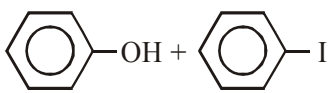
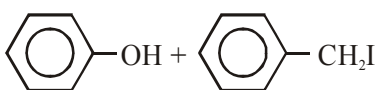
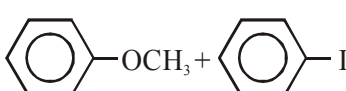
74. Which of the following conformers for ethylene glycol is most stable :-

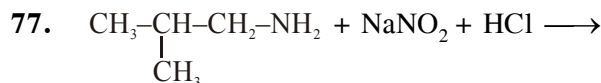


75. Which does not give victor meyer test :-

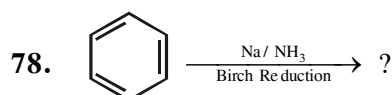
- (1) CH_3OH (2) $\text{CH}_3-\text{C}(\text{CH}_3)_2-\text{OH}$
- (3) $\text{CH}_3-\text{CH}(\text{OH})-\text{CH}_3$ (4) $\text{CH}_3-\text{CH}_2-\text{OH}$



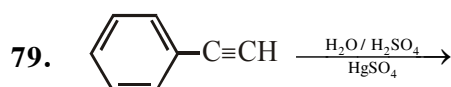
- (1)  (2) 
- (3)  (4) 

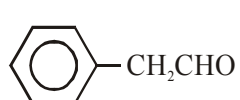
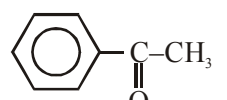
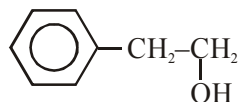
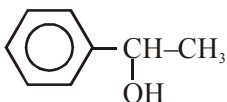


- (1) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_2\text{OH}$ (2) $\text{CH}_3-\underset{\text{CH}_3}{\overset{\text{OH}}{\text{C}}}-\text{CH}_3$
- (3) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{OH}$ (4) $\text{CH}_3-\text{CH}_2-\text{OH}$

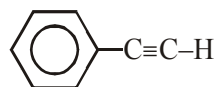
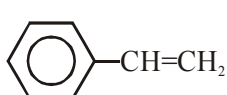


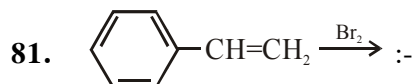
- (1)  (2) 
- (3)  (4) 

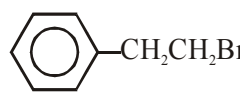
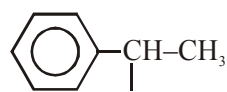
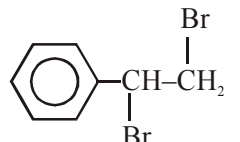


- (1)  (2) 
- (3)  (4) 

80. Which gives white ppt. with Tollen reagent :-

- (1)  (2) $\text{CH}_3-\text{C}\equiv\text{C}-\text{CH}_3$
- (3) $\text{CH}_3-\text{C}\equiv\text{C}-\text{C}_2\text{H}_5$ (4) 



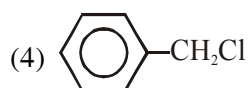
- (1)  (2) 
- (3)  (4) $\phi - \text{Br}$

82. Which gives propane :-

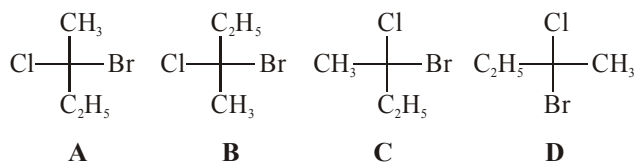
- (1) $\text{CH}_3\text{CH}_2\text{Cl} + \text{CH}_3\text{MgBr}$
- (2) $\text{CH}_3\text{CH}_2\text{MgBr} + \text{CH}_3\text{CH}_2\text{OH}$
- (3) $\text{CH}_3\text{CH}_2\text{COOH} + \text{NaOH} + \text{CaO}/\Delta$
- (4) $\text{CH}_3\text{CH}_2\text{Cl} + \text{Zn} + \text{HCl}$

83. Which shows most difficult N.S.R :-

- (1) $\text{CH}_2=\text{CH}-\text{CH}_2-\text{Cl}$
- (2) CH_3-Cl



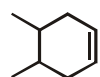
84. Consider the following structures (A), (B), (C) and (D)



Wrong statement will be :-

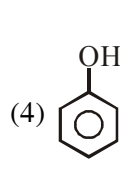
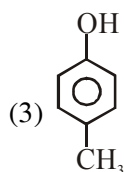
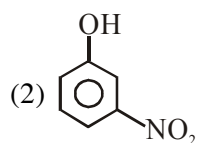
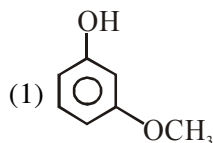
- (1) B and C are identical
- (2) A and B are enantiomers
- (3) A and C are enantiomers
- (4) B and D are enantiomers

85. Arrange the following alkenes in their stability order:-

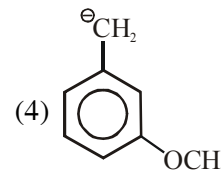
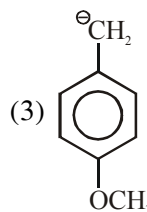
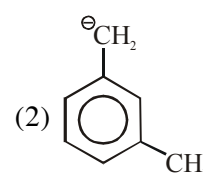
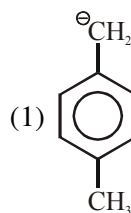


- (1) $A > B > C$
- (2) $C > B > A$
- (3) $B > C > A$
- (4) $C > A > B$

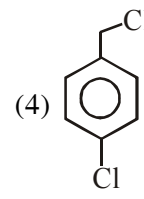
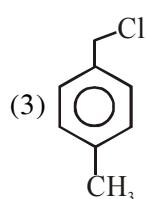
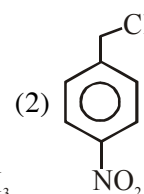
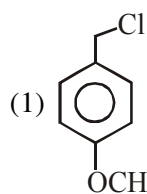
86. Which of the following ring has maximum electron density.



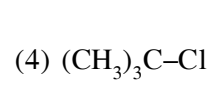
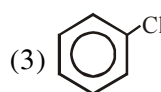
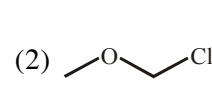
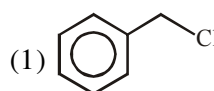
87. Which of the following is most stable carbanion:-



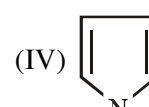
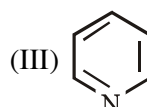
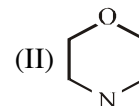
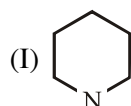
88. Which is most reactive for $\text{S}_\text{N}1$:-



89. Most reactive for $\text{S}_\text{N}1$ reaction is :-



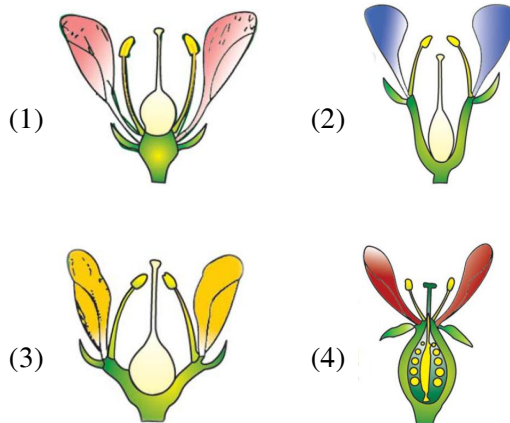
90. Arrange following in their basic strength order:-



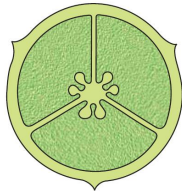
- (1) $\text{IV} > \text{III} > \text{II} > \text{I}$
- (2) $\text{I} > \text{II} > \text{III} > \text{IV}$
- (3) $\text{II} > \text{I} > \text{III} > \text{IV}$

- (4) $\text{II} > \text{I} > \text{IV} > \text{III}$

- 91.** The membrane system of chloroplast is responsible for :
- Trapping the light energy
 - Synthesis of ATP & NADPH
 - Enzymatic reactions for CO_2 incorporation
 - Both (1) and (2)
- 92.** How does PS-II supply electrons continuously :
- by removing from photon
 - by removing from H_2O
 - by removing from CO_2
 - by removing from constituent carotenoids
- 93.** Classification of biosynthetic phase or dark reaction as C_3 & C_4 is primarily based on :
- Initial CO_2 fixation
 - Final CO_2 assimilation
 - First CO_2 receptor
 - Number of ATP get consumed
- 94.** C_4 plants show saturation of photosynthetic reaction at about :-
- 500 ppm of CO_2
 - 360 ppm of CO_2
 - 450 ppm of CO_2
 - 1000 ppm of CO_2
- 95.** C_4 plants consume _____ NADPH molecules to fix 4CO_2 , while C_3 plants requires _____ molecules of NADPH for same amount of CO_2 .
- 8, 8
 - 24, 24
 - 8, 12
 - 12, 8
- 96.** The enzyme which converts glucose to glucose-6 phosphate is :-
- Phosphorylase
 - Glucose - 6-phosphate
 - Hexokinase
 - Glucose synthetase
- 97.** Pyruvate (Pyruvic acid) dehydrogenase is used in converting :-
- Pyruvate to glucose
 - Glucose to pyruvate
 - Pyruvic acid to lactic acid
 - Pyruvate (Pyruvic acid) to acetyl CoA
- 98.** FAD is electron acceptor during oxidation of which of the following :-
- α -keto glutaric acid $\rightarrow$ Succinyl Co-A
 - Succinic acid $\rightarrow$ Fumaric acid
 - Succinyl CoA $\rightarrow$ Succinic acid
 - Fumaric acid $\rightarrow$ Malic acid
- 99.** (A) It is negatively geotropic.
(B) It is positively hydrotropic.
(C) It is positively phototropic.
(D) It is distinguished into nodes and internodes.
(E) It bears apical bud at tip and axillary bud in the axil of leaf.
- Of the above, which statements are correct for stem ?
- A, B, C, E
 - A, C, D, E
 - A, D, E
 - A, D, C, B
- 100.** In Given following stem and their example select right option with suitable example
- Runner – Potato, Zamikand
 - Sucker – Colocasia, Onion
 - Bulb – Turmeric, Ginger
 - Offset – Pistia, Eichhornia
- 101.** In which given following diagram which shows superior ovary



102. According to following diagram select suitable explanation of gynoecium



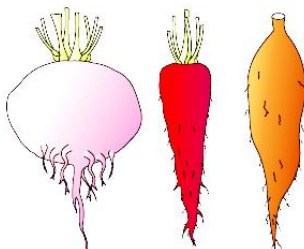
- (1) Trilocular, Tricarpallary, Apocarpus
- (2) Unilocular, Tricarpallary, Syncarpous
- (3) Trilocular, Tricarpallary, Syncarpous
- (4) Trilocular, Monocarpellary, Syncarpous

103. These are the figures representing the modification of :-



- (1) Roots for food storage
- (2) Stem for food storage
- (3) Leaf for food storage
- (4) Inflorescence of plants

104. These roots are of which plants respectively and they are modified for :-



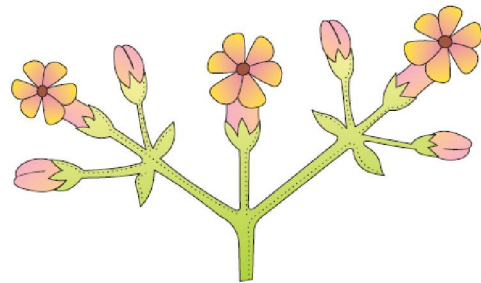
- (1) Turnip, carrot and sweet potato for food storage
- (2) Turnip, carrot and sweet potato for mechanical support
- (3) Turnip, carrot and sweet potato for respiration
- (4) Turnip, orchid and potato for food storage

105. Some plants like *Colchicum*, zaminkand etc. produce corm, the corm is :-

- (1) A swollen root
- (2) A swollen vertical underground stem
- (3) A swollen horizontal underground stem
- (4) A swollen vertical underground root

106. Cyathium and hypanthodium are similar in having

- (1) nectary
- (2) apical pore
- (3) racemose arrangement
- (4) unisexual flowers



107.

Choose incorrect statement regarding the above figure given.

- (a) In this inflorescence main axis terminate into a flower.
 - (b) There is a limited growth.
 - (c) Flowers are borne in a acropetal order.
 - (d) Shown by Jasmine, teak
- (1) b, c (2) only c (3) c, d (4) b, c, d

108. Following figure represents :-



- (1) Hygroscopic roots of orchids
- (2) Respiratory roots of *Begonia*
- (3) Pneumatophores of *Rhizophora*
- (4) Aerial roots of *Monstera*

109. Match the Column-I and Column-II :-

Column-I	Column-II
(A) Pea	(I)
(B) Lemon	(II)
(C) Mustard	(III)
(D) Primrose	(IV)
(E) Sunflower	(V)

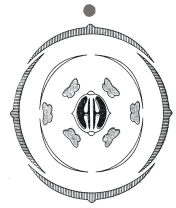
- (1) A - V, B - II, C - I, D - III, E - IV
 (2) A - I, B - II, C - III, D - IV, E - V
 (3) A - V, B - II, C - III, D - I, E - IV
 (4) A - V, B - III, C - II, D - I, E - IV

110. The "Eyes" of the potato tuber are :

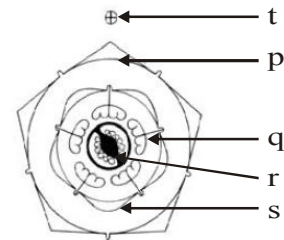
- (1) Root buds
 (2) Flower buds
 (3) Shoot bud
 (4) Axillary buds

111. Which one of the floral formula is correct for given floral diagram ?

- (1) $\oplus \overline{\text{K}}_{(5)} \text{C}_{1+2+(2)} \text{A}_{(9+1)} \underline{\text{G}}_{(2)}$
 (2) $\% \overline{\text{K}}_{(5)} \text{C}_{1+2+(2)} \text{A}_{(9+1)} \underline{\text{G}}_{(2)}$
 (3) $\% \overline{\text{K}}_2 \text{C}_4 \text{A}_{2+4} \underline{\text{G}}_{(2)}$
 (4) $\oplus \overline{\text{K}}_{2+2} \text{C}_4 \text{A}_{2+4} \underline{\text{G}}_{(2)}$



112. Name in given floral diagram :



- (1) P - Calyx, q - Corolla, r - Androecium, s - Gynoecium, t - Mother axis
 (2) P - Calyx, q - Androecium, r - Gynoecium, s - Corolla, t - Mother axis
 (3) P - Corolla, q - Calyx, r - Androecium, s - Gynoecium, t - Mother axis
 (4) P - Corolla, q - Calyx, r - Gynoecium, s - Androecium, t - mother axis

113. Jay, observe the female flower and notice following character.

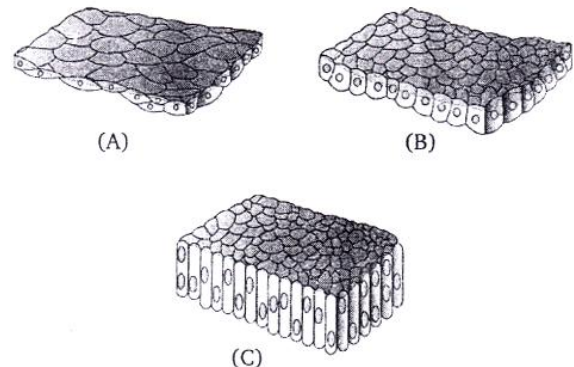
- Bract present, Zygomorphic
 - Unfused, 5 sepals
 - 5 petals fused
 - Inferior, Bicarpellary, Syncarpous ovary
- Choose correct floral formula on the basis of Jay's observation.

- (1) $\text{Br} \oplus \overline{\text{K}}_{(5)} \text{C}_{(5)} \text{A}_1 \underline{\text{G}}_{(2)}$
 (2) $\text{Br} \% \overline{\text{K}}_5 \text{C}_{(5)} \text{A}_0 \overline{\text{G}}_{(2)}$
 (3) $\text{Br} \oplus \overline{\text{K}}_5 \text{C}_{(5)} \text{A}_0 \underline{\text{G}}_{(2)}$
 (4) $\text{Br} \% \overline{\text{K}}_5 \text{C}_5 \text{A}_1 \overline{\text{G}}_{(2)}$

114. Which of the following is mismatched :-

- (1) PASHMINA : Goat
 (2) ANGOORA : Rabbit
 (3) SHAHTOOSH : Chiru
 (4) PASHMINA : Sheep

- 115.** Which of the following method of breeding increases the chance of homozygosity:-
 (1) In- breeding (2) Out- breeding
 (3) Cross- breeding (4) Out- crossing
- 116.** 'Hisardale' is developed by a cross between :-
 (1) Male donkey and female horse
 (2) Sahiwal and Nageri
 (3) Bikaneri ewes and Marino rams
 (4) Jafferabandi and Murrah
- 117.** MOET (multiple ovulation embryo transfer technology) induces which of the following in cow :-
 (1) Implantation
 (2) Follicular maturation
 (3) Super ovulation
 (4) Both 2 and 3
- 118.** What is the main aims of animal breeding in animal husbandry :-
 (1) Increasing the yield of animals
 (2) Increasing the number of animals
 (3) Improving the desirable qualities of the produce
 (4) Both (1) and (3)
- 119.** How many fishes in the list given below are marine ?
 Catla, Pomfret, Common carp, Hilsa, Rohu, Bombay duck, Sardine
 (1) Six (2) Three (3) Four (4) Five
- 120.** Mark the **incorrectly** matched pair ?
 (1) Lemur and spotted cuscus-*convergent evolution*
 (2) Flippers of penguins and dolphins-*Analogous organs*
 (3) Tiger Cat, Koala, Kangaroo, Numbat etc-*Adaptive radiation*
 (4) Industrial melanism [peppered moth]-*stabilising selection*
- 121.** Darwin had made his sea voyage through
 (1) H.M.S. Beagle (2) Titanic
 (3) Carpathia (4) Rocky Mountain
- 122.** Random genetic drift in a population probably results from :-
 (1) high genetically variable individuals
 (2) when migration of a section of population occurs
 (3) constant low mutation rate
 (4) large population size
- 123.** Formation of protobiont in medium is a way of :-
 (1) Getting isolation in medium
 (2) Acquiring energy from sun
 (3) Forming organic compound
 (4) All of the above
- 124.** Choose the correct statement for epithelial tissue :
 (1) Possesses loosely packed cells with little intercellular matrix
 (2) The free surface of epithelium faces the body fluid but not the outside environment
 (3) Has a basal surface and a free surface
 (4) This tissue is always made of only one layer of cells
- 125.** Hardy Weinberg equilibrium is deviated by :-
 (1) Large population (2) Mutation
 (3) Migration (4) Both 2 and 3
- 126.** The tubular parts of nephrons of kidneys and ducts of glands have :
 (1) Cuboidal epithelium
 (2) Columnar epithelium
 (3) Compound epithelium
 (4) Squamous epithelium
- 127.** Identify the diagrams of epithelial tissue alongwith their correct location :



Options	Tissue	Location
1	A = Columnar epithelium	Lining of stomach and intestine
2	B = Cuboidal epithelium	Ducts of glands
3	A = Columnar ciliated epithelium	Lining of stomach and intestine
4	C = Squamous epithelium	Walls of blood vessels

128. In which of the following tissues, the cells are compactly packed with little intercellular matrix ?

- (1) Nervous tissue (2) Connective tissue
(3) Muscular tissue (4) Epithelial tissue

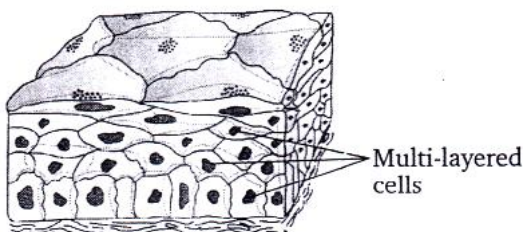
129. Read the following statements w.r.t. epithelial tissue :

- (i) Germinal epithelium produces gametes
(ii) Blood vessels are commonly absent in epithelial tissues
(iii) Epithelial tissue is the last tissue to evolve during evolution
(iv) This tissue is formed first in the embryo in comparison to other tissues
(v) This tissue arise from all the three germ layers i.e., ectoderm, mesoderm and endoderm of embryo

Which of the above statement(s) is/are incorrect ?

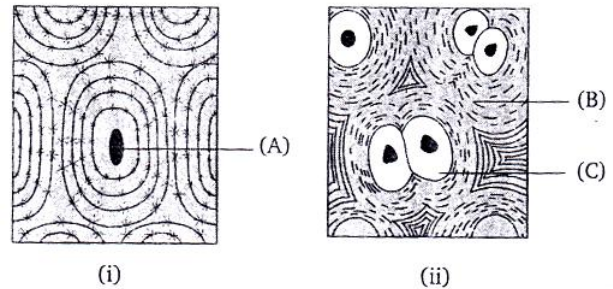
- (1) (ii), (ii) (2) (i), (iii)
(3) (iii) (4) (i), (iv), (v)

130. Identify the epithelium given below and the concerned location :



- (1) Simple squamous ; walls of blood vessels
(2) Simple squamous ; pharynx
(3) Compound columnar ciliated ; buccal cavity
(4) Compound epithelium ; Inner lining of buccal cavity

131. Identify the connective tissues labelled as (i) and (ii) alongwith the parts labelled as A, B and C.



	(i)	(ii)	A	B	C
1	Cartilage	Collagen fibres	Bone	Haversian canal	Chondrocyte
2	Cartilage	Bone	Haversian canal	Collagen fibres	Chondrocyte
3	Bone	Cartilage	Haversian canal	Collagen fibres	Chondrocyte
4	Bone	Cartilage	Chondrocyte	Haversian canal	Collagen fibres

132. Which of the following group of animals show adaptive radiation ?

- (1) Scorpion, elephant, human
(2) Cheetah, Kangaroo, mole
(3) Praying mantis, seal, python
(4) Cockroach, squirrel, toad

133. Select the correct statement.

- (1) Darwinian variations are small and directionless.
(2) Mutations are random and directional.
(3) Fitness is the end result of the ability to adapt and get selected by the nature.
(4) All mammals except whales and camels have seven cervical vertebrae.

134. Industrial melanism as found in peppered moth proves that :

- (1) Melanic forms has no selective advantage in industrial area.
(2) Lighter forms has no selective advantage in polluted and non-polluted industrial area.
(3) Melanism is pollution generated feature.
(4) True black melanic form develops by recurring random mutation.

135. The two key concepts of Darwinian Theory of Evolution are :

- (i) Fitness
- (ii) Branching descent
- (iii) Natural selection
- (iv) Variation

- (1) i & iv
- (2) i & ii
- (3) ii & iii
- (4) All of the above

136. Gene pool of a population tends to remain stable if the population is large, without large scale mutations, without migration and with

- (1) Random mating
- (2) Moderate environmental changes
- (3) Natural selection
- (4) No predator control

137. Which of the following is not true ?

- (1) About 15 mya, *Dryopithecus* and *Ramapithecus* were found.
- (2) *Ramapithecus* was more Ape like
- (3) *Homo erectus* has a large brain around 900 cm³
- (4) The brain capacities were between 650 to 800 cm³ for *homo habilis*.

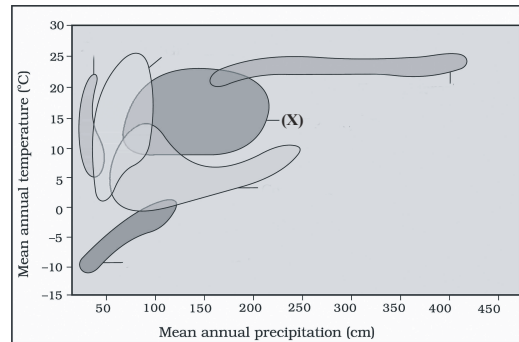
138. Fish die in water polluted by sewage due to :-

- (1) Pathogen
- (2) Foul smell
- (3) Clogging of gills by slit
- (4) Reduction in O₂

139. Match the Column-I to Column-II

- | Column-I | Column-II |
|----------------------------|------------|
| (A) Forest Act | (i) 1974 |
| (B) Water pollution Act | (ii) 1981 |
| (C) National forest policy | (iii) 1927 |
| (D) Air pollution Act | (iv) 1988 |
- (1) A-i, B-ii, C-iii, D-iv
 - (2) A-iii, B-i, C-iv, D-ii
 - (3) A-ii, B-i, C-iv, D-iii
 - (4) A-iv, B-i, C-iii, D-ii

140.



Which type of Biome is represented by the 'X' :-

- (1) Tropical forest
- (2) Temperate forest
- (3) Coniferous forest
- (4) Grass land

141. In equation $\frac{dN}{dt} = rN \left[\frac{K-N}{K} \right]$; here K is defined by :-

- (1) Maximum population density achievable under available resources.
- (2) Carrying capacity of habitat
- (3) Both (1) & (2)
- (4) Maximum intrinsic rate of growth

142. Loss of biodiversity in a region may lead to :-

- (1) Decline in plant production
- (2) Lowered resistance to environmental condition such as drought
- (3) Increased variability in certain ecosystem process
- (4) All

143. Sacred groves are example of in situ conservation are found in :-

- (A) Khasi and Jaintia – In Meghalaya
- (B) Aravalli hills – In Rajasthan
- (C) Chanda and Bastar – In Madhya Pradesh
- (D) Western ghats – In Karnataka
- (1) A, B Only
- (2) B, C Only
- (3) A, C Only
- (4) A, B, C, D

144. What will be effect of addition of domestic sewage in water ?

- (1) C.O.D. remain unaffected
- (2) B.O.D. remain unaffected
- (3) Reduce B.O.D.
- (4) Increase B.O.D.

145. How many statements are correct :-

- (a) Automobiles are major cause for atmospheric pollution atleast in metro cities.
- (b) Water which contain 0.1% of impurities is harmful for drinking.
- (c) As organic matter increased BOD decreases.
- (d) High concentration of DDT affect Ca-metabolism in birds.

- (1) One (2) Three (3) Two (4) None

146. Monotreal protocol aims at :-

- (1) Control of CO₂
- (2) Reduction of ozone depleting substances
- (3) Control of water pollution
- (4) Control over global warming

147. Ultraviolet rays causes :-

- (A) Skin cancer
- (B) High blood pressure
- (C) Blood sugar
- (D) DNA damage

- (1) A, B (2) B, C (3) A, D (4) C, D

148. Scrubber is used to remove

- (1) SO₂
- (2) Particulate matter
- (3) NO₂
- (4) Both 1 & 2

149. Slash and burn agriculture commonly called as Jhum cultivation in north-eastern states of India related to :-

- (1) Deforestation
- (2) Afforestation
- (3) Crop rotation
- (4) Dry farming

150. How many statements are correct :-

- (a) Eutrophication is natural aging of a lake by nutrient enrichment of its water.
- (b) Methane is major cause of green house effect
- (c) Ozone is harmful for human in troposphere.
- (d) Thickness of ozone is measured in metre scale.

- (1) Four (2) Three
(3) Two (4) One

151. Sound becomes noise pollution at level

- (1) Above 30 dB
- (2) Above 80 dB
- (3) Above 100 dB
- (4) Above 120 dB

152. Which is part of in situ conservation:-

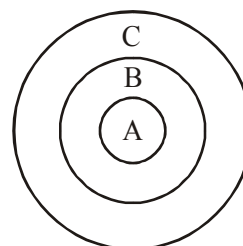
- (1) Gene bank
- (2) Botanical garden
- (3) Tissue culture
- (4) Wild life sanctuaries

153. Match the column-I to column-II.

Column-I		Column-II	
(A)	Amrita Devi	(i)	1980
(B)	JFM	(ii)	1974
(C)	Chipko movement	(iii)	1981
(D)	Air pollution act	(iv)	1731

- (1) A-i ; B-ii ; C-iii ; D-iv
- (2) A-iii ; B-ii ; C-i ; D-iv
- (3) A-iv ; B-iii ; C-ii ; D-i
- (4) A-iv ; B-i ; C-ii ; D-iii

Direction (Q.154 & 155) : Answer the following question by using diagram :



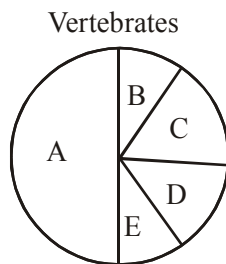
154. Identify the A, B and C area :-

	A	B	C
(1)	Buffer Zone	Transition Zone	Core Zone
(2)	Transition Zone	Buffer Zone	Core Zone
(3)	Core Zone	Buffer Zone	Transition Zone
(4)	Transition Zone	Core Zone	Buffer Zone

155. In which zone no human activities is allowed.

- (1) Core zone
- (2) Buffer zone
- (3) Transition zone
- (4) Core zone and buffer zone

156. Study the given graph and identify the A, B, C, D and E.



	A	B	C	D	E
(1)	Fishes	Mammals	Birds	Reptiles	Amphibians
(2)	Fishes	Reptiles	Amphibians	Birds	Mammals
(3)	Mammals	Fishes	Birds	Amphibians	Reptiles
(4)	Amphibians	Reptiles	Mammals	Fishes	Birds

157. Human derives countless direct economic benefit from nature example food, fire wood, fibre etc. They all are included in :-

- (1) Narrowly utilitarian category
- (2) Broadly utilitarian category
- (3) Ethical category
- (4) None of the above

158. Antrum or follicular cavity present in which of the following structure.

- (1) Primary follicle
- (2) Secondary follicle
- (3) Tertiary follicle
- (4) Atrophied follicle

159. At the time of ovulation ovum first released in which part.

- (1) Body cavity
- (2) Fimbriae of the oviduct
- (3) Ampulla part
- (4) Isthmus part

160. During pregnancy there is

- (1) Continuous ovulation
- (2) Formation of myometrium
- (3) Formation of corpus luteum
- (4) Maintenance of endometrium

161. In spermatogenesis, the phase of maturation involves

- (1) The growth of spermatogonia into primary spermatocyte
- (2) The formation of spermatogonia from gonocytes through mitosis
- (3) The formation of spermatids from primary spermatocytes through meiosis
- (4) The formation of spermatogonia from the spermatocytes through meiosis

162. What would happen if vasa deferentia of man are cut?

- (1) Sperms are non-nucleate
- (2) Spermatogenesis does not occur
- (3) Semen is without sperms
- (4) Sperms are nonmotile

163. Go through the following statements

- (i) Mammary glands are modified sweat glands and each mammary gland consists of 15–25 lobules of the compound tubulo-alveolar type.
- (ii) The tertiary follicle is characterised by a fluid filled cavity called antrum
- (iii) Both LH and FSH attain a peak level towards the middle of the menstrual cycle
- (iv) In oogenesis, meiotic division takes place.

Which of these are correct?

- (1) (ii) & (iii)
- (2) (i), (iii) & (iv)
- (3) (i), (ii) & (iv)
- (4) All are correct

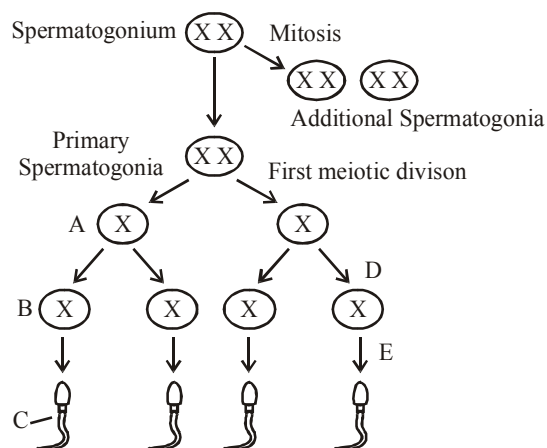
164. The correct sequence of spermatogenic stages leading to the formation of sperms in a mature human testis is

- (1) Spermatid-spermatocyte-spermatogonia-sperms
- (2) Spermatogonia-spermatid-spermatocyte-sperms
- (3) Spermatocyte-spermatogonia-spermatid-sperms
- (4) Spermatogonia-spermatocyte-spermatid-sperms

165. All the copulations not lead to fertilization and pregnancy because.

- (1) Sperm & ovum transported simultaneously.
- (2) Sperm & ovum not transported simultaneously
- (3) Due to acidic nature of uterine medium sperm get damaged.
- (4) All of these

166. Write appropriate term for given A B C D E



A	B	C	D	E
(1) Secondary Spermatocyte	Spermatid	Sperm	Second meiotic Div	Spermio genesis
(2) Secondary Spermatocyte	Spermatid	Sperm	Second meiotic Div	Spermio Matogenesis
(3) Spermatid	Secondary Spermatocyte	Sperm	Spermio genesis	Second meiotic Division
(4) Secondary Spermatocyte	Spermatid	Sperm	No Div	Second meiotic Division

167. Choose an incorrect statement about human reproduction

- (1) Human are sexually reproducing and viviparous.
- (2) Sperm formation ceases in men about the age of seventy year
- (3) Formation of ovum continues in women about the age of fifty years
- (4) Both gametes and ovum are haploid cells produces by glands.

168. It is best for growing crop, it contain 50% sand and 50% silt, it hold maximum water. Which kind of this soil is ?

- (1) Sandy soil
- (2) Loamy soil
- (3) Silt soil
- (4) Laterite soil

169. The term "*Prebiotic soup*" was coined by :-

- (1) Oparin
- (2) Haldane
- (3) Aristotle
- (4) Thales

170. According to the idea of Big bang the universe is :-

- (1) Expanding
- (2) Contracting
- (3) Steady
- (4) Both (2) and (3)

171.  → Birds → Parasites

In above food chain energy transfer from tree to birds and from birds to parasites than what will be the shape of energy pyramid in this food chain ?

- (1) Inverted pyramid
- (2) Upright pyramid
- (3) Spindle shape pyramid
- (4) None of the above

172. The electron flow from PS-II to NADP via ETS and PS-I follows:

- (1) Uphill → Downhill → Uphill → Downhill
- (2) Downhill → Uphill → Downhill → Downhill
- (3) Downhill → Uphill → Uphill → Downhill
- (4) Uphill → Downhill → Downhill → Uphill

173. Match the following column-I with column-II?

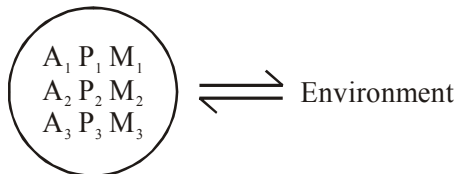
Column - I		Column - II	
(A)	Pyramid of number represents	(i)	Standing crop
(B)	Pyramid of biomass represents	(ii)	Productivity
(C)	Pyramid of energy represents	(iii)	Biotic potential

- (1) A – (ii), B – (iii), C – (i)
 (2) A – (iii), B – (i), C – (ii)
 (3) A – (iii), B – (ii), C – (i)
 (4) A – (ii), B – (i), C – (iii)

174. According to Robert May global species diversity is about :-

- (1) more than 1.5 million
 (2) 7 billion
 (3) 7 million
 (4) 7.5 million

175.



Above given diagram represents a relation with environment. This relation is defined in :

- (1) Species ecology
 (2) Autecology
 (3) Synecology
 (4) Population ecology

176. Mn is required in :

- (1) Chlorophyll-a
 (2) Nucleic acid synthesis
 (3) Plant cell wall
 (4) Photolysis of water during photosynthesis

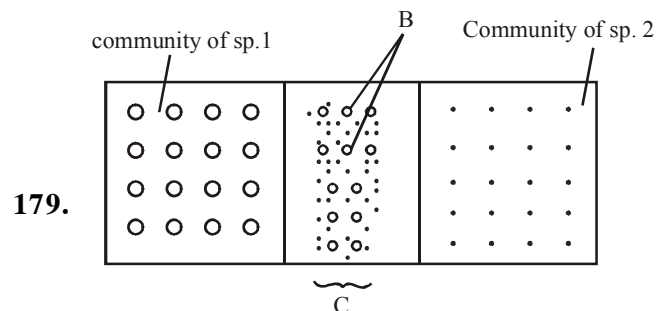
177. Photosynthesis is maximum in :

- (1) Green light
 (2) Blue light followed by red light
 (3) Red light followed by blue light
 (4) Blue light

178. Arrange the events of chemogeny and biogeny given below in correct order :-

- A. Origin of simple inorganic compounds
 B. Origin of complex organic compounds
 C. Origin of living giant molecules
 D. Origin of free atoms
 E. Origin of prebiotic soup
 F. Origin of prokaryotes

- (1) D, A, E, B, C, F
 (2) A, B, E, D, C, F
 (3) D, A, B, C, E, F
 (4) F, C, B, E, A, D



179.

Find out the B and C :-

- (1) B- Ecotone C- edge species
 (2) C- Ecotone B- edge species
 (3) Both are ecotone
 (4) Both are edge effect

180. Black foot disease occurs due to :-

- (1) Cadmium (2) Mercury
 (3) Arsenic (4) Copper